Exercise #8

Introduction To Artificial Intelligence

Illya Starikov

Due Date: April 22nd, 2018

1 State-Space Search Algorithms

Problem Statement. *Give the name of the algorithm that results from each of the following special cases.*

1. Local beam search with $k = 1$. **Steepest-Ascent Hill-Climbing.**
2. Local beam search with one initial state and no limit on the number of states retained. **Breadth First Tree Search.**
3. Simulated annealing with $T = 0$ at all times (and omitting the termination test). **Steepest-Ascent Hill-Climbing.**
4. Simulated annealing with $T = \infty$ at all times. **Random Walk.**
5. Genetic algorithm with population size $N = 1$. **Random Walk.**